

Free Numbers: Local Quaternionic Closure and Global Residual Chart Normalization A Non-Collapsing Quaternionic Chart Algebra

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Abstract

We introduce **free numbers**, a formal framework for representing local algebraic closures and their global residual composition without collapsing all local structures into a single ambient vector space. The construction begins with a local closure F_1 , whose primitive fold plane is two-dimensional and whose completed local algebra is shown to be isomorphic to the real quaternion algebra $\mathbb{H}$. The global structure F is then defined as a finite term system generated from a disjoint family of local closures, quotiented only by admissible equivalences that preserve local algebraic structure, boundary data, conjugation, and residual purity.

The central feature of the framework is the distinction between **admissible transitions**, which allow local charts to be identified, and **residual transitions**, which must not be collapsed. When two local elements cannot be compared by admissible transport, a new generated chart is introduced. This yields a residual chart network in which global composition need not be globally associative, but is instead tracked through finite generation histories.

We further define a Gen-normalization procedure and isolate a minimal rewriting kernel suitable for Lean-style formalization. The kernel includes finite terms, one-step reduction, reflexive-transitive reduction, normal forms, and a well-founded measure for a restricted core system. The purpose of the present paper is not to claim a completed semantic theory, but to provide a precise algebraic and rewriting core for non-collapsing relational state spaces in principle.

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1 Introduction

Many formal models represent a system by embedding all states into a single global space. This is often convenient, but it can erase distinctions between locally valid structures that are not globally comparable. In particular, two local configurations may each possess a coherent internal algebra, while no admissible transition exists that preserves the relevant algebraic, boundary, and residual data between them.

This paper proposes **free numbers** as a formal response to this problem. The guiding idea is simple:

local closure should not imply global collapse.

The construction separates two levels.

First, a local closure F_1 is defined. At this level, the structure closes in a quaternionic form:

$$F_1 \cong_{\mathbb{R}\text{-alg}} \mathbb{H}.$$

Second, a global free-number structure F is constructed from many local closures $F_{1,\alpha}$. These local charts are not identified merely because they are algebraically isomorphic. They may be identified only when an admissible transition preserves the required algebraic and boundary structure. Otherwise, their interaction generates a new chart.

Thus the global structure is not a single quaternion algebra. Rather, it is a finite term system of local quaternionic closures, admissible transports, residual boundaries, and generated charts.

Figure 1 summarizes the central distinction.

2 Informal Motivation

The motivation for free numbers is the need to describe systems in which local algebraic consistency and global comparability are distinct notions.

A local chart may be internally closed. It may have a unit, conjugation, multiplication, and a norm-like self-consistency trace. However, two such charts may still fail to be globally

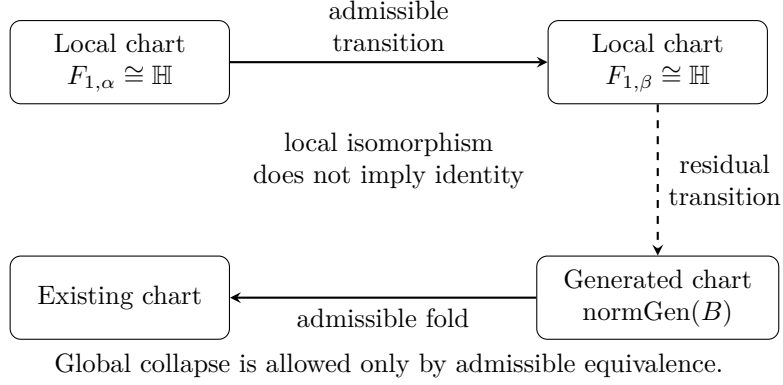


Figure 1: Local closure does not imply global collapse. A solid arrow denotes an admissible transition as described in Section 7; a dashed arrow denotes a residual transition as described in Section 8. The generated chart is normalized rather than collapsed, reflecting the Gen-normalization procedure of Section 11.

equivalent. Their difference may not be an error to be removed, but a residual structure to be preserved.

In this sense, free numbers are designed to represent relational configurations where:

1. local closure is meaningful;
2. global identification is not automatic;
3. residual mismatch must be tracked;
4. new comparison charts may be generated by composition;
5. normalization should not erase non-admissible differences.

The framework is intentionally conservative. It does not assert that all residual structures have a single semantic interpretation. It only provides a formal language in which residual comparison failure is not immediately collapsed.

3 Contributions

The main contributions are as follows.

1. **Local quaternionic closure.** We define a local closure F_1 and identify the algebraic form under which it is isomorphic to the real quaternion algebra $\mathbb{H}$.
2. **Global residual chart construction.** We define a global free-number structure F as a finite term system generated from a disjoint family of local closures:

$$F = \text{Term}_{\mathcal{R}} \left(\bigsqcup_{\alpha \in \mathcal{B}_0} F_{1,\alpha} \right) / \equiv_{\text{adm}} .$$

3. **Admissible versus residual transitions.** We distinguish transitions that preserve the required local structure from transitions whose residuals must be retained.
4. **Generated charts.** We introduce a chart-generation operation for the interaction of locally closed but globally non-comparable elements.

5. **Gen-normalization.** We formulate a normalization scheme that either folds generated charts into existing admissible charts or marks them as normalized residual charts.
6. **Minimal formal kernel.** We provide a Lean-style rewriting core consisting of terms, reductions, reflexive-transitive reductions, normal forms, and a well-founded measure for a restricted system.

4 Scope and Non-Claims

This paper does not claim that free numbers constitute a completed theory of meaning, time, cognition, or physical reality.

The present work is limited to the following claim:

free numbers provide a formal algebraic and rewriting framework for local closures connected by admissible a

In particular, this paper does not assume:

1. that all global products are associative;
2. that all algebraically isomorphic local charts are identical;
3. that residual transitions may be quotiented away;
4. that scalar projections determine equality;
5. that the local quaternionic form exhausts the global structure.

The global structure F is not identified with $\mathbb{H}$. Only the local closure F_1 is quaternionic.

The current Lean mini-kernel uses `Nat` tags for `Chart` and `LocalId`. These tags are provisional syntactic labels only. In particular, chart distinctness, disjoint-copy semantics, and boundary isomorphism are not yet enforced at the kernel level. The kernel proves termination of the restricted Gen-normalization core only; it does not yet formalize the full non-identical-copy semantics of

$$\bigsqcup_{\alpha} F_{1,\alpha}.$$

These components are deferred to later versions under chart-distinctness refinement, `BoundaryIso`, and `EqAdm`.

5 Local Closure F_1

We now describe the local closure F_1 . The purpose of this section is to isolate the local algebraic core before introducing global residual chart generation.

Let U be a two-dimensional real vector space equipped with a negative definite quadratic form. We call U the **primitive fold plane**. Choose an ordered orthonormal basis

$$u, v \in U$$

such that

$$u^2 = v^2 = -e_0,$$

where e_0 is the distinguished local unit.

We impose the anti-commutation relation

$$uv = -vu.$$

Define

$$I := u, \quad J := v, \quad K := uv.$$

Then

$$I^2 = J^2 = -e_0.$$

Moreover,

$$K^2 = (IJ)(IJ) = I(JI)J = I(-IJ)J = -I^2 J^2 = -e_0.$$

The multiplication table is therefore

$$IJ = K, \quad JK = I, \quad KI = J,$$

and

$$JI = -K, \quad KJ = -I, \quad IK = -J.$$

Thus the local closure is

$$F_1 = \text{span}_{\mathbb{R}}\{e_0, I, J, K\}.$$

The three pure directions I, J, K should not be confused with three independent primitive generators. The primitive fold plane is two-dimensional:

$$\dim U = 2.$$

The third pure direction arises from the product of the two primitive directions:

$$K = IJ.$$

Equivalently, K may be regarded as the exterior bivector component associated with

$$\Lambda^2 U.$$

Since

$$\dim \Lambda^2 U = \frac{2(2-1)}{2} = 1,$$

the pure local direction space has dimension

$$\dim(U \oplus \Lambda^2 U) = 2 + 1 = 3.$$

Hence

$$F_1 = \mathbb{R}e_0 \oplus U \oplus \Lambda^2 U$$

has dimension four.

Proposition 5.1 (Local quaternionic form). *The local closure F_1 is isomorphic, as a real algebra, to the quaternion algebra $\mathbb{H}$.*

Proof. Define a real-linear map

$$\Phi : \mathbb{H} \rightarrow F_1$$

by

$$\Phi(1) = e_0, \quad \Phi(i) = I, \quad \Phi(j) = J, \quad \Phi(k) = K.$$

The quaternion multiplication rules are

$$i^2 = j^2 = k^2 = -1,$$

$$ij = k, \quad jk = i, \quad ki = j,$$

with reversed products acquiring a minus sign.

By construction, I, J, K satisfy the corresponding relations:

$$I^2 = J^2 = K^2 = -e_0,$$

$$IJ = K, \quad JK = I, \quad KI = J,$$

and

$$JI = -K, \quad KJ = -I, \quad IK = -J.$$

Therefore Φ preserves multiplication. Since both $\mathbb{H}$ and F_1 have real bases of cardinality four, Φ is a linear bijection. Hence

$$F_1 \cong_{\mathbb{R}\text{-alg}} \mathbb{H}.$$

□

Corollary 5.2 (Local but not global quaternionicity). *The local closure F_1 has quaternionic algebraic form. However, this does not imply that the global free-number structure F is itself a quaternion algebra.*

In the global construction, many local copies $F_{1,\alpha}$ may each be quaternionic while remaining non-identical unless connected by admissible transitions. Thus quaternionicity is a local closure property, not a global collapse principle.

6 Local Charts and Non-Identical Copies

Let $\mathcal{B}_0$ be a family of initial local chart labels. For each $\alpha \in \mathcal{B}_0$, let

$$F_{1,\alpha} = \text{span}_{\mathbb{R}}\{e_{0,\alpha}, I_\alpha, J_\alpha, K_\alpha\}.$$

Each chart satisfies

$$F_{1,\alpha} \cong_{\mathbb{R}\text{-alg}} \mathbb{H}.$$

However, for $\alpha \neq \beta$, we do not identify $F_{1,\alpha}$ and $F_{1,\beta}$ merely because both are quaternionic. Instead, we begin with the disjoint union

$$\tilde{F} = \bigsqcup_{\alpha \in \mathcal{B}_0} F_{1,\alpha}.$$

Thus two elements with the same coordinate expression in two different charts remain distinct unless an admissible equivalence is explicitly given.

7 Admissible Transitions

A transition

$$T_{\beta\alpha} : F_{1,\alpha} \rightarrow F_{1,\beta}$$

is called **admissible** if it preserves the required local structure.

At minimum, an admissible transition must preserve:

1. the distinguished local unit;
2. local multiplication;
3. conjugation;
4. the local self-consistency trace;
5. boundary data associated with chart generation.

If such a transition exists, then elements may be identified under admissible equivalence:

$$x_\alpha \equiv_{\text{adm}} T_{\beta\alpha}(x_\alpha).$$

The admissible equivalence relation is generated only by such structure-preserving transitions and by local algebraic laws.

Residual transitions are not included in admissible equivalence.

Operational criterion for admissibility

In the present version, admissibility is treated as a structural condition rather than as a fully implemented decision procedure. A transition

$$T_{\beta\alpha} : F_{1,\alpha} \rightarrow F_{1,\beta}$$

is admissible only if it preserves all data required to regard the two local charts as the same chart for the purposes of the current construction.

Concretely, the intended admissibility criterion consists of the following preservation requirements:

1. **unit preservation:**

$$T_{\beta\alpha}(e_{0,\alpha}) = e_{0,\beta};$$

2. **multiplication preservation:**

$$T_{\beta\alpha}(xy) = T_{\beta\alpha}(x)T_{\beta\alpha}(y);$$

3. **conjugation preservation:**

$$T_{\beta\alpha}(x^*) = T_{\beta\alpha}(x)^*;$$

4. **local trace preservation:**

$$Q_\beta(T_{\beta\alpha}(x)) = Q_\alpha(x);$$

5. **boundary preservation**, meaning that generated boundary data is mapped to boundary data with the same structural origin.

The abstract predicate `ExistingBoundary` in the minimal Lean kernel should be read as a placeholder for the case where these preservation requirements have already been satisfied at the boundary level. It is not intended to identify charts merely because they are algebraically isomorphic.

A full decision procedure for admissibility, including boundary isomorphism and residual purity, is deferred to later versions.

8 Residual Transitions

A transition that fails to preserve the required data is called **residual**.

A residual transition is not an error. It is a record that two local closures cannot be collapsed without losing structure.

For a candidate transition $S_{\beta\alpha}$, one may define residual components such as

$$\Delta_e^S, \quad \Delta^S, \quad \Delta_*^S, \quad \Delta_Q^S.$$

These respectively measure failure to preserve the unit, multiplication, conjugation, and local quadratic trace.

The collection

$$\text{Res}(S_{\beta\alpha}) = (\Delta_e^S, \Delta_\cdot^S, \Delta_\ast^S, \Delta_Q^S)$$

is called the residual signature of the candidate transition.

Only when the residual signature is pure, or can be reduced to a pure admissible transition by an explicitly allowed normalization, may the corresponding transition contribute to admissible equivalence.

9 Global Free Numbers

The global free-number structure is defined by finite generated terms modulo admissible equivalence:

$$F = \text{Term}_{\mathcal{R}} \left(\bigsqcup_{\alpha \in \mathcal{B}_0} F_{1,\alpha} \right) / \equiv_{\text{adm}} .$$

The term grammar includes:

1. local atomic elements;
2. formal products;
3. conjugates;
4. generated chart terms.

Products are syntactic unless they occur inside a common local chart or can be reduced by admissible transport. In particular, global associativity is not assumed.

Thus $(XY)Z$ and $X(YZ)$ may represent different generation histories unless an admissible equivalence relates them.

10 Generated Charts

Given elements

$$x_\alpha \in F_{1,\alpha}, \quad y_\beta \in F_{1,\beta},$$

if no admissible comparison allows them to be multiplied inside a common existing chart, their interaction generates a new chart:

$$\gamma = \text{Gen}(\alpha, x; \beta, y).$$

This operation does not merely create a value. It creates a new local context in which the interaction can be represented.

The boundary of the generated chart records its origin:

$$\partial\gamma = (\alpha, x; \beta, y).$$

Generated charts are therefore not identified solely by algebraic isomorphism. Their boundary data is part of their identity.

11 Gen-Normalization

Gen-normalization determines whether a generated chart can be folded into an existing admissible chart or must be retained as a new normalized residual chart.

The normalization procedure follows four informal rules.

1. If the product is already local, compute locally.
2. If admissible transport places the factors in a common chart, compute there.
3. If the generated chart is admissibly equivalent to an existing chart, fold it into that chart and mark it normalized.
4. Otherwise, retain it as a new normalized generated chart.

The purpose of normalization is not to erase residuals. It is to prevent repeated unresolved generation while preserving non-admissible differences.

12 Minimal Rewriting Kernel

A minimal rewriting kernel can be isolated as follows.

Terms are generated by

`atom, mul, star, gen, normGen.`

A one-step reduction relation *Red* includes:

1. local multiplication;
2. folding a generated chart into an existing normalized chart;
3. marking a fresh generated chart as normalized.

In the restricted kernel, pure transport is temporarily excluded from *Red* until a transport cost, a transport normality condition, or a separate preprocessing relation is introduced.

Why transport is excluded from the restricted kernel

The relation `PureMove` represents admissible transport between local elements. It is declared in the Lean kernel but intentionally excluded from the restricted reduction relation *Red*.

The reason is technical. A transport step may preserve both

openGenCount

and

termSize.

If such a step were included without an additional decreasing measure, the restricted termination proof would no longer be immediate.

Future versions may restore transport in one of three ways:

1. by assigning a **transport cost** that strictly decreases along admissible transport chains;
2. by requiring a **transport normality** condition, so that transport is performed only as a preprocessing step;
3. by separating transport from *Red* and treating it as an equivalence-side operation rather than as a reduction step.

In v0.2-pre.2, the goal remains deliberately narrower: to isolate a terminating Gen-normalization core before restoring the full transport system.

A lightweight measure is defined by the repeated-addition form

$$weight(X) = openGenCount(X) + openGenCount(X) + termSize(X).$$

This is mathematically equal to

$$2 \cdot openGenCount(X) + termSize(X),$$

but the repeated-addition form matches the Lean kernel exactly and keeps the natural-number expression syntactically minimal. For the restricted kernel, every one-step reduction strictly decreases this measure:

$$Red(X, Y) \Rightarrow weight(Y) < weight(X).$$

This gives a well-founded core suitable for defining normal forms.

13 Normal Forms

Let Red^* denote the reflexive-transitive closure of Red . A term N is a normal form of X if

$$Red^*(X, N)$$

and there is no term Y such that

$$Red(N, Y).$$

Equivalently,

$$NormalForm(N) := \neg \exists Y, Red(N, Y).$$

The minimal kernel does not yet prove uniqueness of normal forms modulo admissible equivalence. That requires weak confluence and the full admissible equivalence relation.

14 Deferred Proof: Weak Confluence

The present version establishes a terminating restricted reduction kernel. In this restricted kernel, pure transport is not included in the one-step reduction relation Red . This allows the Gen-normalization core to be equipped with a simple well-founded measure.

Weak confluence modulo admissible equivalence is not required for the restricted termination result and is deferred to a later version. In the full system, weak confluence must account for the interaction among:

1. local multiplication;
2. admissible transport;
3. existing-chart folding;
4. fresh generated-chart normalization;
5. admissible equivalence of boundaries;
6. residual purity.

The intended statement is that if

$$X \rightarrow Y$$

and

$$X \rightarrow Z,$$

then Y and Z can be reduced to terms Y' and Z' such that

$$Y' \equiv_{\text{adm}} Z'.$$

This proof is deferred because it depends on the full admissible equivalence relation, boundary isomorphism, and residual purity criteria, which are outside the minimal kernel of the present version.

15 Projection and Γ

A scalar projection may be defined on normalized self-consistency traces, but projection must not create equality.

That is,

$$\rho(X) = \rho(Y)$$

must not imply

$$X \equiv_{\text{adm}} Y.$$

A projection suitable for global magnitude comparison must satisfy at least:

1. admissible invariance;
2. local normality;
3. non-constancy;
4. residual sensitivity.

A global magnitude defect may then be expressed as

$$\Gamma_{\rho}^{\text{mag}}(X, Y) = \frac{\rho(N_F^{\text{norm}}(XY))}{\rho(N_F^{\text{norm}}(X))\rho(N_F^{\text{norm}}(Y))}.$$

In local admissible sectors, one expects $\Gamma = 1$. In residual sectors, $\Gamma \neq 1$ may occur.

16 Discussion

Free numbers are designed to formalize a distinction that is often blurred: local closure is not the same as global identity.

The local quaternionic closure F_1 provides a compact algebraic core. The global structure F then preserves the fact that many such local closures may coexist without being collapsible into one global space.

This gives a formal language for non-collapsing relational state spaces. It may be useful wherever comparison failure is structural rather than accidental.

The present construction may also admit possible extensions to residual-based reconstruction. Those topics are outside the scope of the present paper. Here we restrict attention to the algebraic and rewriting core of free numbers.

17 Conclusion

We have introduced free numbers as a formal structure built from local quaternionic closures, admissible transitions, residual signatures, generated charts, and Gen-normalization.

The key point is that local algebraic equivalence does not automatically imply global identification. Residual differences are retained unless an admissible transition justifies their collapse.

The resulting structure is neither a single quaternion algebra nor a conventional global vector space. It is a finite generated chart network with controlled normalization.

A minimal Lean-style rewriting kernel supports the claim that the restricted Gen-normalization core can be organized as a terminating reduction system. Future work includes the full formalization of admissible equivalence, residual purity, weak confluence, projection-sensitive invariants, and transport-normalized reductions.

A Minimal Lean Kernel

The following appendix contains the minimal rewriting kernel used to represent terms, one-step reductions, transitive reductions, normal forms, and a decreasing weight measure. This kernel is intentionally restricted: pure transport is not yet included in the reduction relation.

The termination theorem is proved relative to Lean’s standard `propext` axiom together with the abstract relations `LocalMul`, `ExistingBoundary`, and `FreshBoundary`. Running

```
#print axioms FreeNumKernel.red_weight_decreases
```

is expected to list exactly these dependencies:

```
propext
FreeNumKernel.ExistingBoundary
FreeNumKernel.FreshBoundary
FreeNumKernel.LocalMul
```

The abstract relation `PureMove` is declared but excluded from `Red`, and is therefore not expected to appear in this dependency list.

```
/-
FreeNumKernel / Minimal.lean
Free Numbers Lean minimal kernel v0.2-pre.2

Scope:
- restricted rewriting core only
- no mathlib dependency intended
- pure transport declared but excluded from Red

Axiom and deferred component list for v0.2-pre.2.

A1. LocalMul
  Abstract local multiplication relation.
  Intended later replacement:
    concrete multiplication table for local quaternionic closure.

A2. ExistingBoundary
  Abstract relation indicating that a generated boundary can fold
  into an existing normalized boundary.

A3. FreshBoundary
  Abstract relation indicating that a generated boundary cannot fold
  into an existing boundary and must be retained as normGen.
```

```

A4. PureMove
  Abstract admissible transport relation.
  Declared in v0.2-pre.2 but excluded from Red.
  It will be restored only after adding either:
    - transportCost, or
    - TransportNormal, or
    - a preprocessing relation separate from Red.

A5. BoundaryIso
  Deferred boundary isomorphism relation.

A6. ResidualPurity
  Deferred residual purity criterion.

A7. EqAdm
  Deferred admissible equivalence relation.

A8. WeakConfluence
  Deferred weak confluence modulo EqAdm.

A9. EffectiveMagProjection
  Deferred admissible, non-constant, residual-sensitive projection.

A10. TransportNormalizedReduction
  Deferred restoration of admissible transport into reduction.

The restricted kernel proves only:
  Red X Y -> weight Y < weight X

This establishes termination for the restricted Gen-normalization core,
not for the full admissible transport system.
-/

namespace FreeNumKernel

/-- Chart identifier.
Provisional Nat representation.

Important:
  this minimal kernel uses Nat only as a syntactic chart label.
  It does not yet enforce chart distinctness, disjoint-copy semantics,
  or boundary isomorphism at the kernel level.
  These are deferred to later versions.
-/
abbrev Chart := Nat

/-- Local element identifier. Provisional Nat representation. -/
abbrev LocalId := Nat

/-- Generated chart boundary. -/
structure Boundary where
  leftChart : Chart
  leftId : LocalId
  rightChart : Chart
  rightId : LocalId
deriving Repr, DecidableEq

/-- Local element inside a chart. -/

```

```

structure LocalElem where
  chart : Chart
  id : LocalId
deriving Repr, DecidableEq

/--
Finite generated terms.

atom : local element
mul : syntactic product; global associativity is not assumed
star : conjugation / return operation
gen : unresolved generated chart boundary
normGen : normalized generated chart boundary
-/
inductive Term where
  | atom : LocalElem -> Term
  | mul : Term -> Term -> Term
  | star : Term -> Term
  | gen : Boundary -> Term
  | normGen : Boundary -> Term
deriving Repr, DecidableEq

/-- Count open, unresolved generated chart nodes. -/
def openGenCount : Term -> Nat
  | Term.atom _ => 0
  | Term.mul t1 t2 => openGenCount t1 + openGenCount t2
  | Term.star t => openGenCount t
  | Term.gen _ => 1
  | Term.normGen _ => 0

/-- Syntactic tree size. -/
def termSize : Term -> Nat
  | Term.atom _ => 1
  | Term.mul t1 t2 => 1 + termSize t1 + termSize t2
  | Term.star t => 1 + termSize t
  | Term.gen _ => 1
  | Term.normGen _ => 1

/-- Restricted termination weight.

The paper uses the same repeated-addition form:

  openGenCount t + openGenCount t + termSize t

This is mathematically equal to:

  2 * openGenCount t + termSize t

but the repeated-addition form is used here to keep the restricted
Nat expression syntactically minimal.
-/
def weight (t : Term) : Nat :=
  openGenCount t + openGenCount t + termSize t

/-- Abstract local multiplication relation. -/
axiom LocalMul : LocalElem -> LocalElem -> LocalElem -> Prop

/-- Abstract admissible transport, declared but excluded from Red in v0.2-pre.2. -/

```

```

axiom PureMove : LocalElem -> LocalElem -> LocalElem -> Prop

/-- Abstract relation for folding a generated boundary into an existing boundary. -/
axiom ExistingBoundary : Boundary -> Boundary -> Prop

/-- Abstract relation for retaining a fresh generated boundary as normalized. -/
axiom FreshBoundary : Boundary -> Prop

/--
Restricted one-step reduction.

G1_local:
  local atom multiplication

G3_existing:
  generated boundary folds into an existing normalized boundary

G4_mark:
  fresh generated boundary is retained as normalized

Pure transport is intentionally excluded from Red in v0.2-pre.2.
-/
inductive Red : Term -> Term -> Prop where
  | G1_local {x y z : LocalElem} :
    LocalMul x y z ->
    Red
    (Term.mul (Term.atom x) (Term.atom y))
    (Term.atom z)

  | G3_existing {B B' : Boundary} :
    ExistingBoundary B B' ->
    Red
    (Term.gen B)
    (Term.normGen B')

  | G4_mark {B : Boundary} :
    FreshBoundary B ->
    Red
    (Term.gen B)
    (Term.normGen B)

/-- Red seen in the decreasing direction for well-founded recursion. -/
def RedRel : Term -> Term -> Prop :=
  fun Y X => Red X Y

/-- Reflexive-transitive closure of Red. -/
inductive RedStar : Term -> Term -> Prop where
  | refl {X : Term} :
    RedStar X X

  | step {X Y Z : Term} :
    Red X Y ->
    RedStar Y Z ->
    RedStar X Z

/-- A normal form is a term with no outgoing Red step. -/
def NormalForm (X : Term) : Prop :=
  Not (Exists (fun Y : Term => Red X Y))

```

```

/-- N is a normal form reached from X. -/
def IsNFOf (X N : Term) : Prop :=
  RedStar X N NormalForm N

/-- Existing-boundary fold decreases open generated nodes. -/
theorem red_G3_decreases
  {B B' : Boundary} (_h : ExistingBoundary B B') :
  openGenCount (Term.normGen B') < openGenCount (Term.gen B) := by
  simp [openGenCount]

/-- Fresh-boundary normalization decreases open generated nodes. -/
theorem red_G4_decreases
  {B : Boundary} (_h : FreshBoundary B) :
  openGenCount (Term.normGen B) < openGenCount (Term.gen B) := by
  simp [openGenCount]

/-- Restricted Red never increases open generated nodes. -/
theorem red_openGen_noninc
  {X Y : Term} (h : Red X Y) :
  openGenCount Y <= openGenCount X := by
  cases h <;> simp [openGenCount]

/-- Restricted Red strictly decreases the termination weight. -/
theorem red_weight_decreases
  {X Y : Term} (h : Red X Y) :
  weight Y < weight X := by
  cases h <;> simp [weight, openGenCount, termSize]

/-- Same decrease expressed in the RedRel direction. -/
theorem redRel_weight_decreases
  {X Y : Term} (h : RedRel Y X) :
  weight Y < weight X := by
  exact red_weight_decreases h

end FreeNumKernel

```

B Axioms and Deferred Components

The present version separates proved components, definitional components, and deferred components. Table 1 lists the abstract assumptions used in the minimal formal kernel and indicates their intended role in later versions. The dependency profile of `red_weight_decreases` is explicitly exposed rather than hidden: it is relative to `propext` and to the abstract relations A1–A3.

The minimal kernel proves only the following restricted claim:

$$\text{Red}(X, Y) \Rightarrow \text{weight}(Y) < \text{weight}(X).$$

This establishes termination for the restricted Gen-normalization core, not for the full admissible transport system. It is not a proof of weak confluence, global normal-form uniqueness, full admissible transport, or full chart-distinctness semantics.

Label	Component	Role	Status
A1	<code>LocalMul</code>	Local atom product	Abstract relation
A2	<code>ExistingBoundary</code>	Fold into existing boundary	Abstract relation
A3	<code>FreshBoundary</code>	Retain as normalized	Abstract relation
A4	<code>PureMove</code>	Admissible transport	Excluded from Red
A5	<code>BoundaryIso</code>	Boundary equivalence	Deferred
A6	<code>ResidualPurity</code>	Residual purity criterion	Deferred
A7	<code>EqAdm</code>	Admissible equivalence	Deferred
A8	<code>WeakConfluence</code>	Normal form uniqueness modulo <code>EqAdm</code>	Deferred
A9	<code>EffectiveMagProjection</code>	Residual-sensitive scalar projection	Deferred
A10	<code>TransportNormalizedReduction</code>	Restore transport into reduction	Deferred

Table 1: Axioms and deferred components in v0.2-pre.2.